

ASSIGNMENT 4

❖ Network Threats and Mitigation & Physical Security and Risk

Task (1):

Network Threat	Description	Real-World Example in a Popular App/Service
1. Phishing	Deceptive communications (like fraudulent messages or links) designed to trick users into revealing sensitive information, such as login credentials or financial details.	WhatsApp: Attackers sending fake verification messages or impersonating customer support to trick users into handing over their 6-digit registration code, compromising their account.
2. Distributed Denial of Service (DDoS)	Flooding a server, service, or network with a massive volume of traffic from multiple sources to overwhelm it and render it unavailable to legitimate users.	Instagram: Coordinated botnets overwhelming Instagram's servers with fake traffic, causing temporary service outages and preventing normal users from refreshing their feeds or posting.
3. Man-in-the-Middle (MitM)	An attacker secretly intercepts and relays communications between two parties who believe they are directly communicating with each other, often to steal data.	Paytm: An attacker intercepting unencrypted or compromised public Wi-Fi traffic during a transaction to eavesdrop on session tokens or payment requests between the user's device and the payment gateway.
4. Malware	Malicious software (such as viruses, trojans, or spyware) designed to infiltrate devices, damage systems, or covertly extract sensitive information.	WhatsApp: A malicious APK file disguised as a trendy app update or a document shared via chat that, once downloaded, installs spyware to monitor the user's keystrokes or media.
5. Password Attacks	Methods used to gain unauthorized access to user accounts by guessing, cracking, or stealing passwords (e.g., brute-force attacks, credential stuffing).	Instagram: Attackers using automated scripts (credential stuffing) to test leaked username and password combinations from other data breaches to break into user accounts with weak or reused passwords.

Task (2):

```
def brute_force_pin(target_pin):  
    attempts = 0  
  
    print(f"[*] Starting brute-force attack on target  
PIN: {target_pin}\n")  
  
    for i in range(10000):  
        attempts += 1  
        guess = f"{i:04d}"  
  
        if guess == target_pin:  
            print(f"[+] PIN found: {guess}")  
            print(f"[+] Total attempts it took:  
{attempts}")  
            return attempts  
  
    print("[-] PIN not found.")  
    return attempts
```

```
if __name__ == "__main__":  
    target = "7392"  
    brute_force_pin(target)
```

Task (3):

1. Unauthorized Access (Tailgating or Intrusion):

- **The Risk:** Unauthorized individuals gaining entry by following an authorized employee through a door (tailgating) or malicious insiders tampering with hardware. This can lead to data theft, hardware sabotage, or physical tampering.
- **Mitigation:** Implement **mantraps (sally ports)**—a dual-door entry system where the first door must close completely before the second one opens, paired with multi-factor biometric authentication (e.g., retina or fingerprint scanners).

2.Environmental Hazards (Fire and Electrical Outages)

- **The Risk:** Electrical faults or overheating can cause server racks to catch fire. Standard water sprinkler systems can ruin electronics permanently and cause extended service downtime.
- **Mitigation:** Install **clean-agent fire suppression systems** (such as FM-200 or Novec 1230) that use gas to extinguish electrical fires safely without leaving residue or damaging equipment.

3.Climate Control and Cooling Failures:

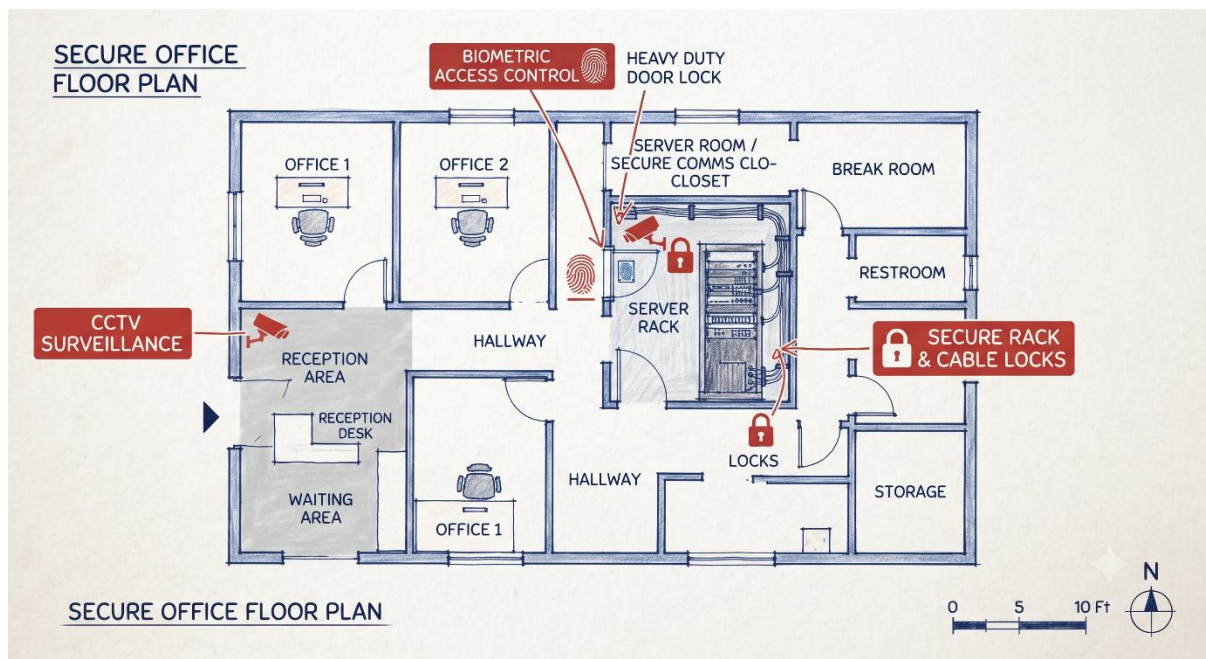
- **The Risk:** High-density servers generate massive amounts of heat.
- **Mitigation:** Deploy redundant HVAC systems (N+1 configuration) backed by real-time automated environmental sensors that trigger instant alerts and failover cooling routines.

4.Power Grid Failures and Surges:

- **The Risk:** Sudden blackouts, brownouts, or major power surges can instantly shut down servers, corrupting active databases (crucial for live food orders or ticket bookings) and destroying power supply units.

- **Mitigation:** Use an enterprise-grade **Uninterruptible Power Supply (UPS)** system for immediate battery backup bridge power, paired with automatically starting diesel generators for long-term outages.

Task (4):



Task (5):

1. Implement Network Segmentation and VLANs:

- **Action:** Isolate lab computers onto a dedicated Virtual Local Area Network (VLAN) separate from the university's core administrative network and public student Wi-Fi.
- **Purpose:** Prevents lateral movement if a lab machine is compromised by malware or a brute-force attack.

2. Deploy State-Preservation Software (Deep Freeze/Reimaging):

- **Action:** Install disk-protection software or automated network imaging tools (like Deep Freeze or FOG) that wipe and restore the system image to a clean baseline every time a workstation reboots.
- **Purpose:** Instantly neutralizes malware, keyloggers, or unauthorized software installed by users during lab sessions.

3.Enforce Port Security and 802.1X Authentication:

- **Action:** Disable unused switch ports in the lab and configure active ports with 802.1X authentication or strict MAC address filtering.
- **Purpose:** Stops students or external actors from plugging unauthorized rogue switches, personal laptops, or malicious network sniffers into live wall jacks.

4.Secure Hardware with Chassis and Cable Locks:

- **Action:** Use heavy-duty steel cable locks to anchor desktop towers and monitors to lab desks, and secure internal chassis covers with physical locks.
- **Purpose:** Deters component theft (such as RAM, GPUs, and processors) in an open-access academic environment.

5.Install Biometric or RFID Access Control and CCTV:

- **Action:** Restrict lab entry using student/faculty ID card scanners or biometric readers, supplemented by wide-angle CCTV cameras monitoring all entry

points and blind spots.

- **Purpose:** Controls unauthorized entry during off-hours and provides an audit trail for incident investigations.